

Modern Topics in Information Technology (MTIT)

Introduction to Prompt Engineering Lab - 2026

Activity 1:

Objective:

By the end of this lab, you will be able to:

- Use Microsoft Copilot to summarize and extract key information from Word documents, Excel spreadsheets, and PowerPoint presentations.
- Apply Copilot prompts to generate executive summaries, highlights, and actionable insights.
- Combine information from multiple documents to create comprehensive reports.
- Evaluate the effectiveness of AI-assisted document analysis and summarization.

Required Files:

Download and save the following files to your OneDrive folder and then open them in the appropriate Microsoft 365 app.

1. **Mystic Spice Premium Chai Market Analysis Presentation.pptx**
2. **Contoso Chai Tea market trends.xlsx**
3. **Market Analysis Report for Mystic Spice Premium Chai Tea.docx**

Exercise 1: Summarizing a Word Document with Copilot

Objective

Learn how to use Copilot in Word to generate summaries and extract key insights from lengthy market analysis reports.

Steps

1. Open the file 'Market Analysis Report for Mystic Spice Premium Chai Tea.docx' in Microsoft Word.
2. Click on the Copilot icon in the toolbar or ribbon.
3. In the Copilot pane, type the following prompt:
"Summarize this document in 5 key points"
4. Review the summary generated. Take note of the main themes identified by Copilot.
5. Try a more specific prompt:
"What are the main market opportunities mentioned in this report?"
6. Compare the responses from both prompts. Note the differences in detail and focus.
7. Ask Copilot to:
"Create an executive summary highlighting the key findings and recommendations"
8. Copy the executive summary into a new document for reference.

Exercise 2: Analyzing Data Trends in Excel with Copilot

Objective

Use Copilot in Excel to analyze market trends data and generate insights from numerical information.

Steps

1. Open 'Contoso Chai Tea market trends.xlsx' in Microsoft Excel.
2. Ensure your data is formatted as an Excel table. If not, select the data range and press Ctrl+T to create a table.
3. Open Copilot in Excel from the toolbar.
4. Try the following prompts one by one:
"Summarize the data in this table"

"What are the top 3 trends shown in this data?"

"Highlight any interesting patterns or insights"
5. Ask Copilot to add a formula column if applicable:
"Add a column to calculate the growth rate"
6. Request a chart suggestion:
"Suggest a chart to visualize these trends"
7. Create the suggested chart and format it for clarity.
8. Document your observations about the data analysis in a new sheet labeled "Analysis Notes".

Exercise 3: Extracting Insights from PowerPoint Presentations

Objective

Use Copilot in PowerPoint to summarize presentation content and create speaker notes.

Steps

1. Open 'Mystic Spice Premium Chai Market Analysis Presentation.pptx' in PowerPoint.
2. Browse through the slides to familiarize yourself with the content.
3. Click on the Copilot icon in PowerPoint.
4. Ask Copilot:
"Summarize this presentation"
5. Review the summary and note how Copilot captures the main points from multiple slides.
6. Try a more targeted prompt:
"What are the key recommendations in this presentation?"
7. Navigate to slide 3 and ask Copilot:
"Generate detailed speaker notes for this slide"
8. Add the generated speaker notes to the slide.
9. Create a FAQ section by asking:
"Create 5 potential questions and answers based on this presentation"
10. Add a new slide at the end titled "FAQ" and insert the generated Q&A content.

Exercise 4: Synthesizing Information from Multiple Documents

Objective

Learn to use Copilot to combine insights from Word, Excel, and PowerPoint files into a comprehensive report.

Steps

1. Create a new Word document.
2. Open Copilot in Word.
3. Use the following prompt to reference all three files:
"Create a comprehensive market analysis report based on /Market Analysis Report for Mystic Spice Premium Chai Tea.docx, /Contoso Chai Tea market trends.xlsx, and /Mystic Spice Premium Chai Market Analysis Presentation.pptx"
4. Review of the generated report. Note how Copilot integrates information from different sources.
5. Ask Copilot to add specific sections:
"Add a section on competitive advantages based on the available data"
"Include a risk analysis section"
6. Format the document with:
A title page

Table of contents

Appropriate headings and subheadings

Page numbers
7. Ask Copilot to create an action plan:
"Based on the analysis, create a prioritized action plan with timelines"

Activity 2:

Objective:

Complete the Google Cloud Skills Boost lab on Generative AI with Vertex AI: Prompt Design to learn fundamental prompt engineering techniques and best practices.

Lab Overview:

In this lab, you will explore prompt design using Vertex AI on Google Cloud. You will learn how to craft effective prompts for large language models and understand key concepts in prompt engineering.

Lab Link:

https://www.skills.google/course_templates/976/labs/594524